

Predicting the number of satellites of female horseshoe crabs

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Introduction

When the nesting season approaches, a female horseshoe crab, together with a male horseshoe crab attached on her back, comes ashore to nest. Non-attached male crabs come ashore as well and compete with other males for fertilization with a female near her nest. Those male crabs are called **satellites**.

Brockmann (1996) analyzed the number of satellites of female crabs in terms of physical variables. The dataset **Horseshoe crab satellite** contains the data analyzed by Brockmann.

In this report, we try to find a model able to predict the distribution of the number of satellites of a female horseshoe crab.

Final model

The chosen model for prediction describes the number of satellites as following a Zero-Inflated Negative Binomial (ZINB) distribution whose parameters depend only on the **weight** of the female horseshoe crab, i.e. a distribution with density function

$$\Pr(Y = j) = \begin{cases} \pi + (1 - \pi)f(0) & \text{if } j = 0 \\ (1 - \pi)f(j) & \text{otherwise} \end{cases}$$

where $f(j)$ is the probability density function of a Negative Binomial (NB) distribution with mean $\mu = \exp(\hat{\beta}_1 + \mathbf{weight} \cdot \hat{\beta}_2)$ and variance $\mu + \mu^2/\hat{\theta}$; and where $\hat{\pi} = \text{logit}^{-1}(\hat{\gamma}_1 + \mathbf{weight} \cdot \hat{\gamma}_2)$ is the proportion of zeros. The coefficients $\hat{\beta}_i$, $\hat{\gamma}_i$ and $\hat{\theta}$ are obtained by Most Likelihood Estimation (MLE) from the data.

The dataset

173

The dataset contains observations of ~~173~~ female horseshoe crabs. The following variables are associated to each female :

- **crab** : This is just a numbering of the female crabs, so we can drop it.
- **sat** : A count variable indicating the number of satellites of the female crab. It goes from 0 to 15, but could potentially go higher.

- **y** : A binary indicator variable taking value 1 if the female has at least 1 satellites and 0 otherwise.
- **weight** : The weight of the female in kg.
- **width** : The shell's width of the female in cm.
- **color** : A factor with ordered levels going from 1 to 4 coding the shell's colour of the female in increasing order of darkness.
- **spine** : A factor with ordered levels 1 : Both good, 2 : One worn or broken, 3 : Both worn or broken, indicating the damage state of the two spines of the female.

First, we drop the **y** column. When fitting a Negative Binomial model (with log link) with all covariates, including **y**, the standard error of the associated coefficient explodes with order of magnitude 10^5 . The column **sat** should give us the same and more information than **y**.

Then, looking at the correlation of numeric variables in Table 1, we see that **weight** and **width** are highly linearly correlated ($\text{cor} = 0.89$). Most likely, we will only need to keep one of those two variables.

Finally, we notice that there are 62 females out of 173 (~36.42 %) with zero male satellites. This seems quite large so we will look in more detail for overdispersion and zero-inflation.

Table 1 – Correlation table of numeric variables

	sat	weight	width
sat	1.000	0.369	0.340
weight	0.369	1.000	0.887
width	0.340	0.887	1.000

Overdispersion and Zero-Inflation assessment

We first fit a Poisson model and a Negative Binomial model with all covariates and we investigate for overdispersion or zero-inflation. Recall that a Poisson distribution is the restriction of a Negative Binomial model for the dispersion $\alpha = 1/\theta = 0$. To assess whether the data shows overdispersion or not, we can perform a Likelihood Ratio Test for the null hypothesis $H_0 : \alpha = 0$ and the alternative $H_1 : \alpha > 0$. The statistics is, as usual, given by

$$-2(\mathcal{L}(\text{poisson}) - \mathcal{L}(\text{negative binomial})),$$

where $\mathcal{L}(\cdot)$ is the fitted log-likelihood of the model, and the null is rejected if the statistics is large enough. The asymptotic distribution of the statistics under the null has a probability mass of one-half at zero and a half- $\chi^2(1)$ distribution above zero because the alternative is one-sided (see Cameron 2013). Here, the statistics is 159.56 with critical value 2.70 at level 5%, so the null is strongly rejected. This suggests overdispersion and lack of fit of a Poisson model.

To test for zero-inflation, people often make use of the Vuong test for non-nested models between a Negative Binomial model and its zero-inflated counterpart, but Wilson (2015)

suggests that this use of the test is wrong. Instead, he advises to plot the individual contributions to the likelihood of the observed data under the zero-inflated model hypothesis and the non-zero inflated model one against each other. If zero-inflation is not present, the points should lie approximately along the line $x = y$. The plot, shown in Figure 1, suggests that the data is zero-inflated.

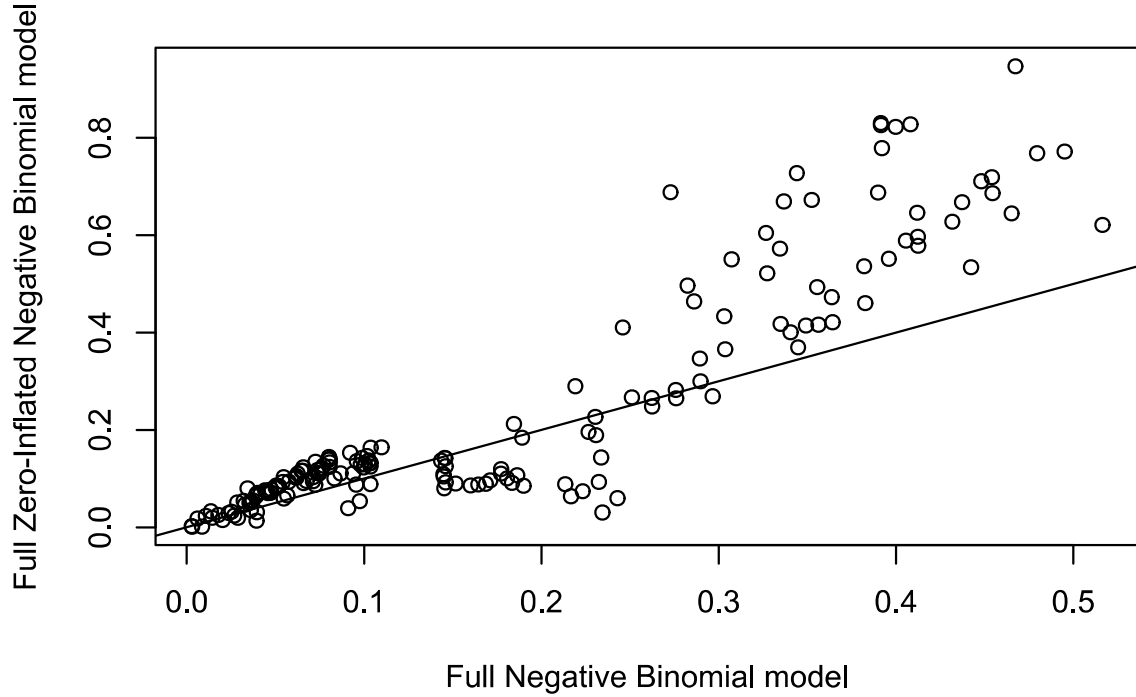


Figure 1 – Individual contributions to the likelihood

Model Selection

From the two above observations, a Negative Binomial model or a Zero-Inflated Negative Binomial model seems more suitable than a Poisson model. Therefore, we will carry out our model selection for those two types of model (with log link only, but models with other link functions could be explored in a similar fashion). The model selection is performed by :

1. Looking at individual Wald Tests for significant covariates and try reducing full models with Likelihood Ratio Tests ;
2. Performing a best subset selection with respect to AIC and BIC ;
3. Cross-validating the models from Step 1 and Step 2.

We consider the full NB model again. The individual Wald Tests shown in Table 2 for the full NB model suggest that the only significant covariate at level 0.05 is **weight**.

A Likelihood Ratio Test (with statistics $-2(\mathcal{L}(\text{simpler model}) - \mathcal{L}(\text{more complex model}))$) between the nested models $\text{sat} \sim \text{weight}$ and $\text{sat} \sim \text{weight} + \text{width} + \text{color} + \text{spine}$, i.e. for the null hypothesis that every coefficient β_i except the one for **weight** is zero, largely fails to reject the null at level 5% as shown in Table 3. Note that in this case, the alternative is two-sided and the asymptotic distribution of the statistics is chi-squared by Wilk's theorem.

Table 2 – Negative Binomial model $\text{sat} \sim .$ with log link and dispersion parameter $\hat{\theta} = 0.96$

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-0.718	1.877	-0.383	0.702	
weight	0.701	0.356	1.966	0.049	*
width	-0.002	0.100	-0.024	0.980	
color.L	-0.450	0.322	-1.398	0.162	
color.Q	0.169	0.247	0.685	0.493	
color.C	0.055	0.178	0.311	0.756	
spine.L	0.030	0.176	0.172	0.863	
spine.Q	0.216	0.294	0.733	0.463	

Table 3 – Likelihood Ratio Test between two nested NB models

Model	2 x log-lik.	Test	df	LR stat.	Pr(Chi)
weight	-748.6				
weight + width + color + spine	-745.3	1 vs 2	6	3.323	0.767

For the full ZINB model, the individual Wald Tests (not shown here) suggest that the only significant covariates at level 5% are **weight** and **color**.

Again, we carry out a Likelihood Ratio Test between a full ZINB model and the simpler model $\text{sat} \sim \text{weight} \mid \text{weight}$. We fail to reject the null at level 5% with a **p-value** of 0.07. However, when testing between the nested models $\text{sat} \sim \text{weight} \mid \text{weight}$ and the simpler model $\text{sat} \sim \text{weight} \mid 1$, the null is strongly rejected with a **p-value** of order of magnitude 10^{-7} .

Thus, there is no evidence going against the adequacy of the NB model $\text{sat} \sim \text{weight}$ or the ZINB model $\text{sat} \sim \text{weight} \mid \text{weight}$.

Now, we perform a best subset selection on NB models with all possible covariates and two-way interaction terms, where an interaction term can be added only when the associated lower order terms are also present in the model, using AIC and BIC. As seen in Table 4, the model minimizing both AIC and BIC is $\text{sat} \sim \text{weight}$. The other best models almost always have only **weight** or **width** as covariates. This is not surprising since those two variables are highly correlated, so those models should be approximately the same.

Table 4 – Best NB models for AIC and BIC

Best NB models for BIC	BIC	Best NB models for AIC	AIC
sat ~ weight	764.1	sat ~ weight	754.6
sat ~ width	766.8	sat ~ weight + width + weight :width	755.5
sat ~ weight + width	769.2	sat ~ weight + width	756.6
sat ~ weight + width + weight :width	771.2	sat ~ width	757.3
sat ~ weight + spine	773.9	sat ~ weight + color	757.9

We repeat the above best subset selection (without interaction terms this time) on ZINB models. The result is shown in Table 5. We can see that the models tend to be more complex but have lower AIC and BIC than NB models.

To find the best model for prediction among the above models, we perform a 10-fold cross-validation, repeated 100 times, with logarithmic scoring rule.

The result of the cross-validation process shown in Appendix A reveals that the NB model `sat ~ weight` is the best model for prediction among the above NB models. However, all the NB models perform worse than the ZINB models. The simple addition of a constant zeros' proportion $\hat{\pi}$ to the best NB model `sat ~ weight` improves the cross-validation mean from 2.173 to 2.139.

The best cross-validation mean is 2.059, achieved by the ZINB model `sat ~ weight + color | width + color`. This is not a large improvement compared to the 2.061 mean of the ZINB model `sat ~ weight | weight`.

Notice finally that the ZINB models selected by AIC all perform better than the models selected by BIC and all have a cross-validation mean of ≈ 2.060 .

Summary of the model selection step

The above analysis showed that the NB model `sat ~ weight` is the best with respect to BIC and AIC among NB models and the best with respect to predictive performance among the considered NB models. The ZINB model `sat ~ weight | weight`, although not optimal for BIC or AIC (BIC = 731.3, AIC = 715.5), has almost the best predictive performance among the considered NB and ZINB models and is simpler than most of them. We have also seen that those two models could not be rejected by a Likelihood Ratio Test at level 5%.

Before deciding which model to choose, we will assess how well the models preferred by cross-validation fit our data. We have already seen that the data shows signs of zero-inflation, so it's more likely that a ZINB model fits the data better than a NB model.

Table 5 – Best ZINB models for AIC and BIC

Best ZINB models for BIC	BIC	Best ZINB models for AIC	AIC
sat ~ weight + width + color weight + width + color + spine	701.7	sat ~ weight + width + color width + color	711.6
sat ~ weight + width + color + spine width + color + spine	701.9	sat ~ weight + color width + color	712.3
sat ~ weight + width + color + spine weight + width + color	702.2	sat ~ weight width + color	713.1
sat ~ weight + width + color width + color + spine	702.3	sat ~ weight + width width + color	713.1
sat ~ weight + width + color + spine width + color	702.6	sat ~ weight + color weight + width + color	713.4

Assessment of fit

First, we study and compare the fit of the Zero-Inflated Negative Binomial model $\text{sat} \sim \text{weight} \mid \text{weight}$ and the Negative Binomial model $\text{sat} \sim \text{weight}$. For comparison purposes, we will also consider the Poisson model (with log link) $\text{sat} \sim \text{weight}$ and the Zero-Inflated Poisson model $\text{sat} \sim \text{weight} \mid \text{weight}$. We will proceed as follow :

1. Compare the expected frequencies (under a specific model) and the observed frequencies with visual plots ;
2. Analyze randomized quantile residuals with a normal QQ-plot.

Note that since a zero-inflated model is not a true Generalized Linear Model (GLM), deviance is not defined for those type of models. Consequently, deviance residuals and pseudo R-squared measures are also not defined for zero-inflated models. Besides AIC and BIC, which were already used in the model selection process, there is no standard measure of goodness-of-fit to compare zero-inflated models and non zero-inflated models.

As stated, we first compare how the expected frequencies $\text{exp}_j = \sum_{i=1}^n f(j; \hat{\theta}, \hat{\mu}_j, \hat{\pi}_j)$, $j = 1, \dots, n$, of each model differ from the observed frequencies $\text{obs}_j = \sum_{i=1}^n I_{y_i=j}(y_i)$ with a hanging rootogram plot.

More precisely, given a model, for each $j = 1, \dots, n$, we plot $\sqrt{\text{exp}_j}$ with a bar hanging from $\sqrt{\text{exp}_j}$ to $\sqrt{\text{exp}_j} - \sqrt{\text{obs}_j}$. If the model fits the data well, the bars should go from $\sqrt{\text{exp}_j}$ to approximately 0. Figure 2 shows the four rootogram plots, with the ± 1 warning limit from Tukey (1972) in blue.

The plots reveal that the Poisson and the Negative Binomial model underfit the expected number of zeros and overfit small count values of the response, which, again, suggests that the data is zero-inflated. Both zero-inflated model fit almost perfectly the expected number of zeros. However, the Zero-Inflated Poisson largely underfit the number of 1.

Finally, the Zero-Inflated Negative Binomial model has an overall better fit than the three other models and does not display a clear deviation between the expected and observed frequencies for most count values.

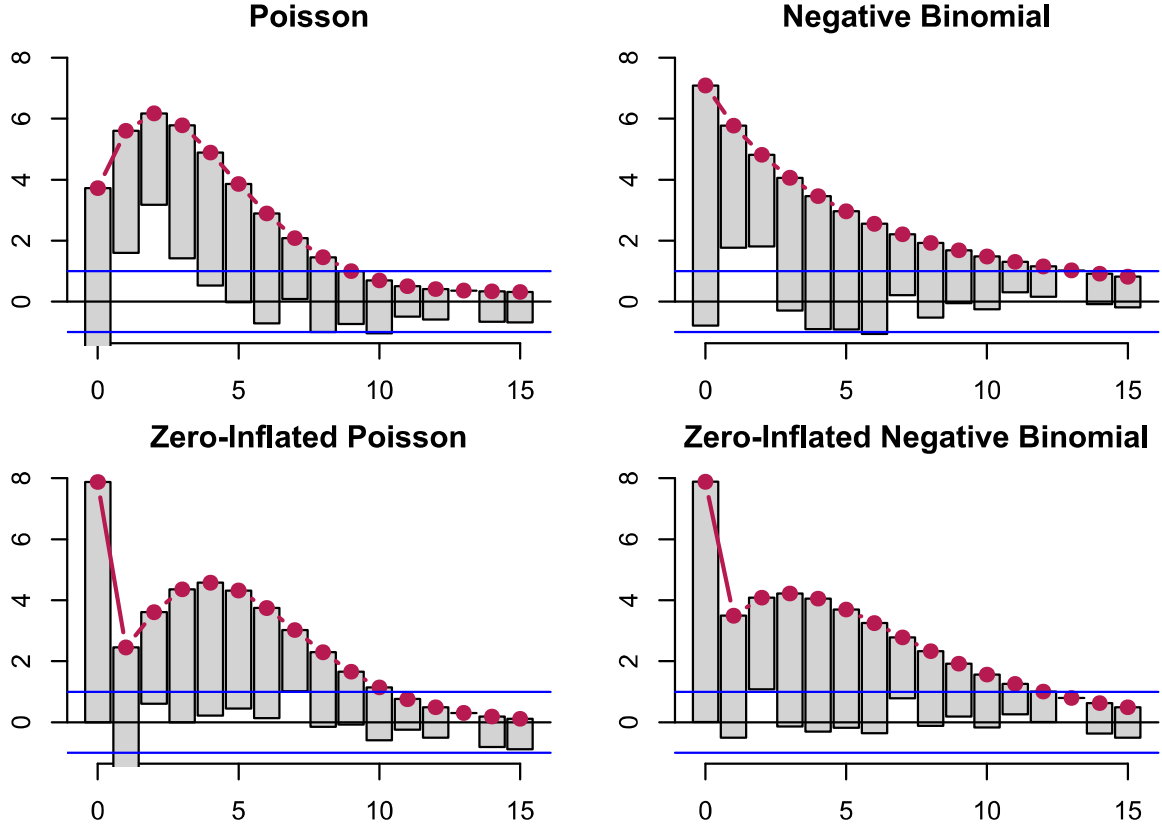


Figure 2 – Comparison of rootogram plots for various models.

x-axis : Number of satellites

y-axis : $\sqrt{\text{Expected Frequencies}}$

In the GLM context, one can look at multiple type of residuals : Pearson, Deviance, Standardized Pearson, Anscombe, ... but they are not all defined for zero-inflated models. Even so, the main issue with those residuals is that normality is not to be expected from them in most cases. For this reason, we will look at randomized quantile residuals, which follow a standard normal distribution under the hypothesis that the model fits the data (see Dunn and Smyth 1996). This assumption can be checked with a normal QQ-plot as shown in Figure 3. If the model fits the data, the points should lie along the line $x = y$.

The normality assumption seems to hold well for the NB and ZINB model. The QQ-plot of the Zero-Inflated Poisson model might indicate some right-skewness in the distribution of the residuals but it is not so apparent. However, the Poisson model fails to meet the normality assumption : the QQ-plot shows that the residuals come from a distribution which starts as Gaussian with negative mean on the left and is clearly right-skewed.

The above analysis, and in particular the rootogram plots, showed that the ZINB model $\text{sat} \sim \text{weight} \mid \text{weight}$ seems to fit the data better the Negative Binomial model $\text{sat} \sim \text{weight}$.

One can also look at the rootogram plots and normal QQ-plots of the other ZINB models selected by cross-validation in the previous step (which are the ZINB models with best AIC), but there will be no clear improvement in the fit (see Appendix B).

This analysis, together with the model selection step, justifies our choice for the final model :

1. A Negative Binomial model is needed to deal with overdispersion ;
2. A correction for the zero count is needed because of zero-inflation ;
3. The ZINB model $\text{sat} \sim \text{weight} \mid \text{weight}$ cannot be rejected by Likelihood Ratio Test at level 0.05, has good predictive performance and fits the data well ;
4. ZINB models selected by AIC meet the same criteria 2, but the model $\text{sat} \sim \text{weight} \mid \text{weight}$ is simpler and there was no clear difference between the performance and fit of those models.

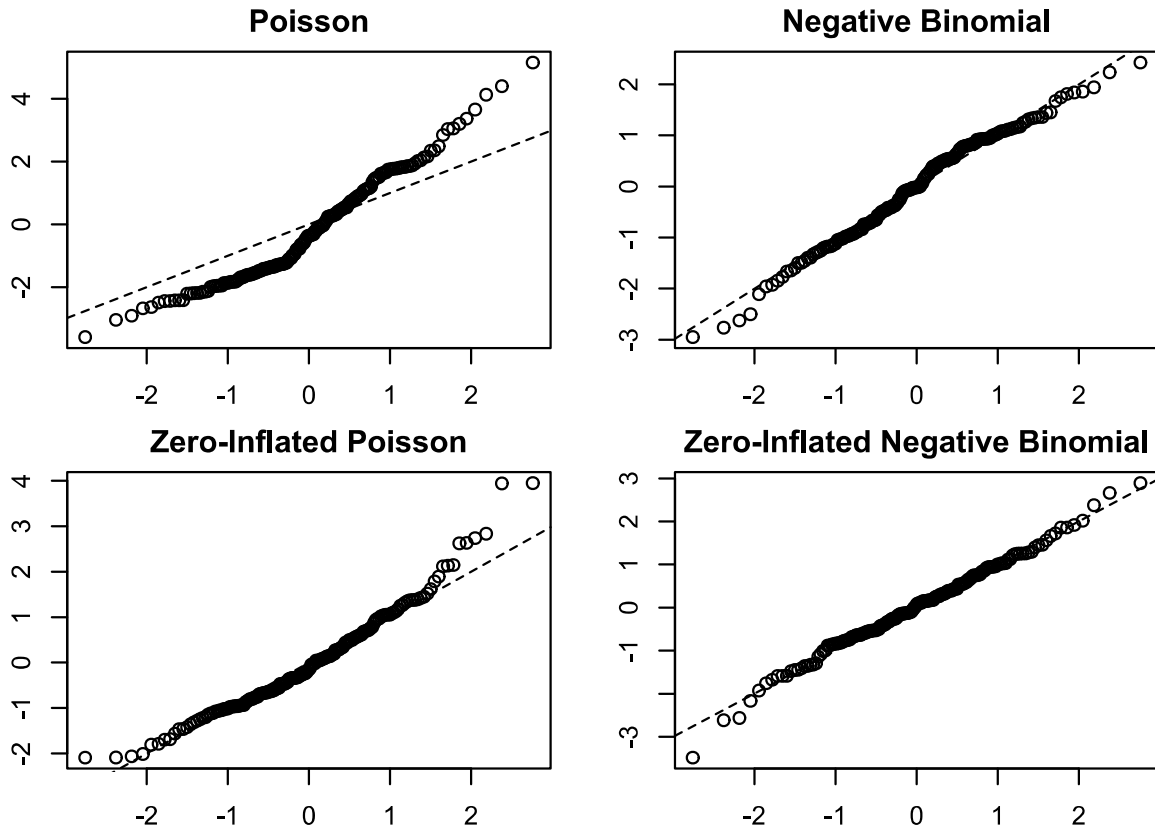


Figure 3 – Normal QQ-plot of randomized quantile residuals for various models.

x-axis : Theoretical quantiles

y-axis : Empirical quantiles

Table 6 – Count component

	Estimate	Std. Error
(Intercept)	0.898	0.305
weight	0.217	0.112
Log(theta)	1.601	0.355

Table 7 – Zero component

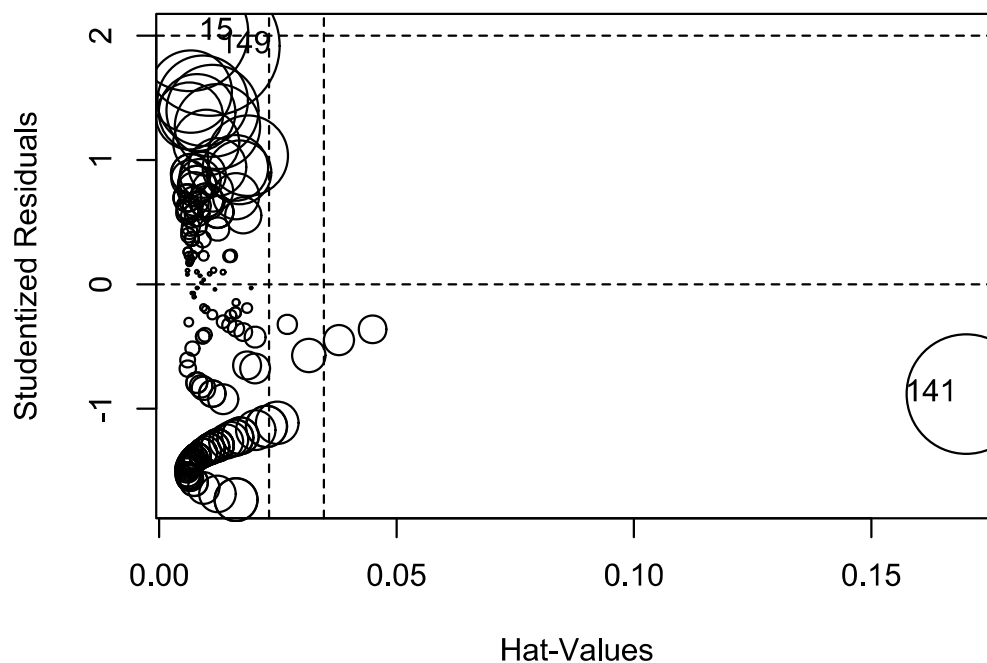
	Estimate	Std. Error
(Intercept)	3.756	0.984
weight	-1.913	0.432

Leverage points and outliers

Tables 6 and 7 show the coefficients and standard errors for the fitted ZINB model `sat ~ weight | weight`. Wald Tests and p-values are not shown because the model selection step invalidates them.

Since a zero-inflated model is not a true GLM, there is no standard method to detect influential points. Instead, we will look at diagnostic plots for the NB model `sat ~ weight` and see if the suggested fixes are also relevant for the ZINB model.

Figure 4 plots the Standardized Pearson Residuals (also called Studentized Residuals) against the diagonal of the GLM hat matrix. The radius of a circle is proportional to the Cook's distance of the associated observation.

Figure 4 – Influence plot for NB model `sat ~ weight`

The plot indicates that observations 15 and 149 are outliers and observation 141 is a leverage

point. Looking at the dataset, we see that observation 15 is the only observation with 14 satellites recorded and weights 2.30 kg; observation 149 has 10 satellites and weights 1.90 kg. This is quite smaller than the mean of the weights for observations with ≥ 7 satellites, which is 2.87. Observation 141 is the heaviest horseshoe crab observed in this dataset; it weights 5.20 kg, which is 1.35 kg more than the second heaviest crab in the dataset. This might explain why they are influential points.

Table 8 shows the fitted ZINB model $\text{sat} \sim \text{weight} \mid \text{weight}$ when we remove some combinations of those observations. We see that removing those observations doesn't tend to change anything except increase the coefficient for **weight** in the count component of the regression, which makes sense from our previous look at the data. This is not so important for us. We still keep all the observations; uncommon observations can happen, so there is no legitimate reason to get rid of them here.

	- Obs. 141	- Obs. 15	- Obs 149	- Obs. 15, 149
Count model : (Intercept)	0.869 (0.346)	0.828 (0.295)	0.800 (0.308)	0.727 (0.297)
Count model : weight	0.228 (0.129)	0.237 (0.108)	0.249 (0.112)	0.270 (0.108)
Count model : Log(theta)	1.563 (0.353)	1.775 (0.393)	1.655 (0.367)	1.853 (0.413)
Zero model : (Intercept)	3.750 (0.990)	3.729 (0.970)	3.822 (0.991)	3.794 (0.976)
Zero model : weight	-1.913 (0.435)	-1.890 (0.424)	-1.934 (0.434)	-1.910 (0.425)
Num. obs.	172	172	172	171

Standard errors in parentheses

Table 8 – Fitted ZINB models $\text{sat} \sim \text{weight} \mid \text{weight}$ when some observations are removed

Conclusion

We consider the coefficients of the ZINB model $\text{sat} \sim \text{weight} \mid \text{weight}$ from Tables 6 and 7.

Given the weight of a female horseshoe crab, the probability density function for the number of satellites Y of the female is

$$\Pr(Y = j) = \begin{cases} \pi + (1 - \pi)\Pr(X = 0) & \text{if } j = 0 \\ (1 - \pi)\Pr(X = j) & \text{otherwise} \end{cases}$$

where

$$\begin{aligned} X &\sim \text{NegBin}(\mathbb{E} = \mu, \text{Var} = \mu + \mu^2/\theta), \\ \mu &= \exp(0.90 + 0.22 \cdot \text{weight}), \quad \theta = 4.96, \\ \pi &= \frac{\exp(3.76 - 1.91 \cdot \text{weight})}{1 + \exp(3.76 - 1.91 \cdot \text{weight})} \end{aligned}$$

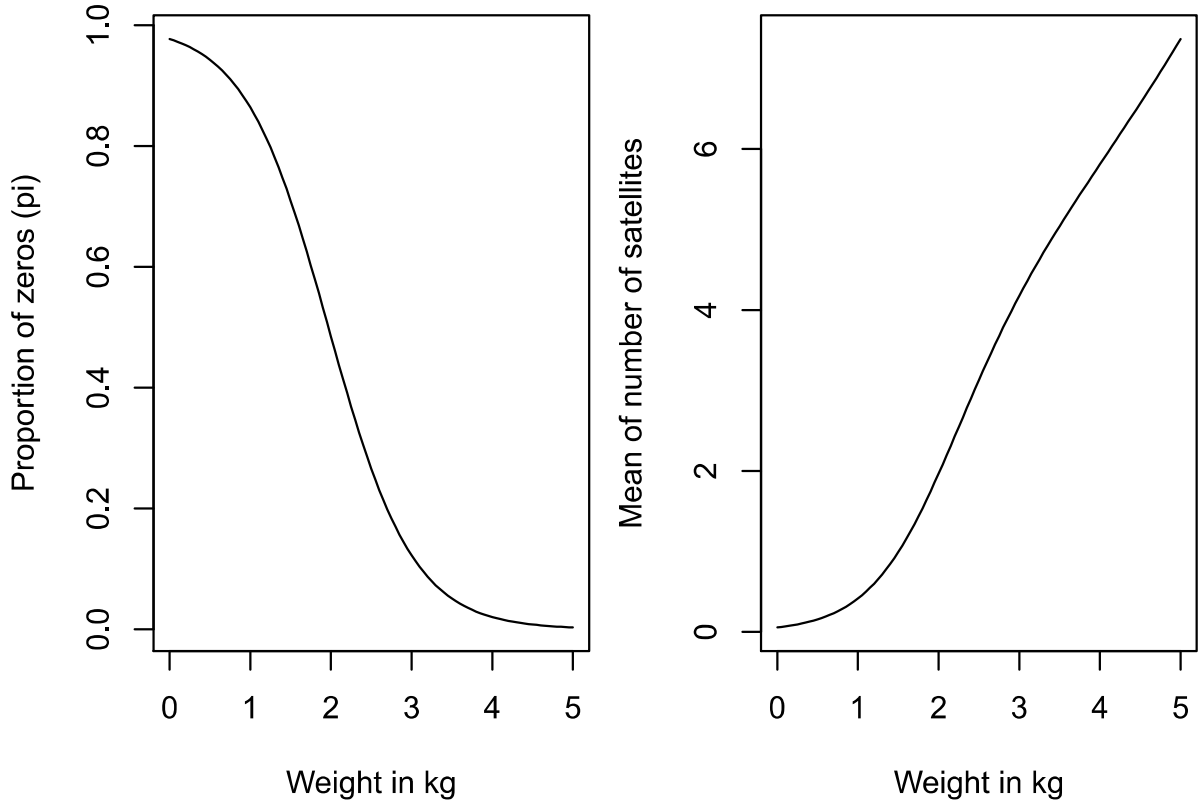


Figure 5 – Behaviour of $\mathbb{E}(Y)$ and π as weight increases

The mean of Y is given by $(1 - \pi)\mu$ and this could be used with care as a prediction for the number of satellites. Standard errors should be taken into account, but they are not sufficient to provide confidence intervals. Confidence intervals could be obtained using Bootstrapping methods, but we do not consider them here.

The coefficients of the model are meaningful. An increase of 1 kg in weight increases the mean of the Negative Binomial part from μ_0 to $\mu_1 = \exp(0.22) \cdot \mu_0 = 1.25 \cdot \mu_0$ and decreases the proportion of zeros as shown in Figure 5. Therefore, an increase of 1 kg in weight always increases the mean of Y from $(1 - \pi_0)\mu_0$ to $(1 - \pi_1)\mu_1 \geq 1.25 \cdot (1 - \pi_0)\mu_0$.

Furthermore, the proportion of zeros has more influence on the number of satellites when the weight of the crab is small and becomes negligible (≤ 0.1) when the crab is heavy (≥ 3 kg). The probability of having zero satellite increases as the weight decreases according to our model.

Finally, we should keep in mind that the model was built from the data of a particular population of horseshoe crabs, so predictions for crabs from other populations might not be accurate and we should take this into account.

Appendix A

Table 9 – 10-fold cross-validation on best NB, ZINB models for AIC, BIC

Model	Cross-validation mean	Cross-validation mean's standard deviation
sat ~ .	2.204	0.011
sat ~ weight	2.173	0.003
sat ~ width	2.180	0.003
sat ~ weight + width	2.177	0.004
sat ~ weight + width + weight :width	2.175	0.006
sat ~ weight + spine	2.184	0.006
sat ~ weight + color	2.187	0.008
sat ~ . .	2.091	0.016
sat ~ weight + width + color weight + width + color + spine	2.077	0.015
sat ~ weight + width + color + spine width + color + spine	2.088	0.015
sat ~ weight + width + color + spine weight + width + color	2.079	0.014
sat ~ weight + width + color width + color + spine	2.073	0.014
sat ~ weight + width + color + spine width + color	2.075	0.014
sat ~ weight + width + color width + color	2.060	0.013
sat ~ weight + color width + color	2.059	0.011
sat ~ weight width + color	2.060	0.009
sat ~ weight + width width + color	2.061	0.011
sat ~ weight + color weight + width + color	2.063	0.011
sat ~ weight weight	2.061	0.006
sat ~ weight 1	2.134	0.005

Appendix B

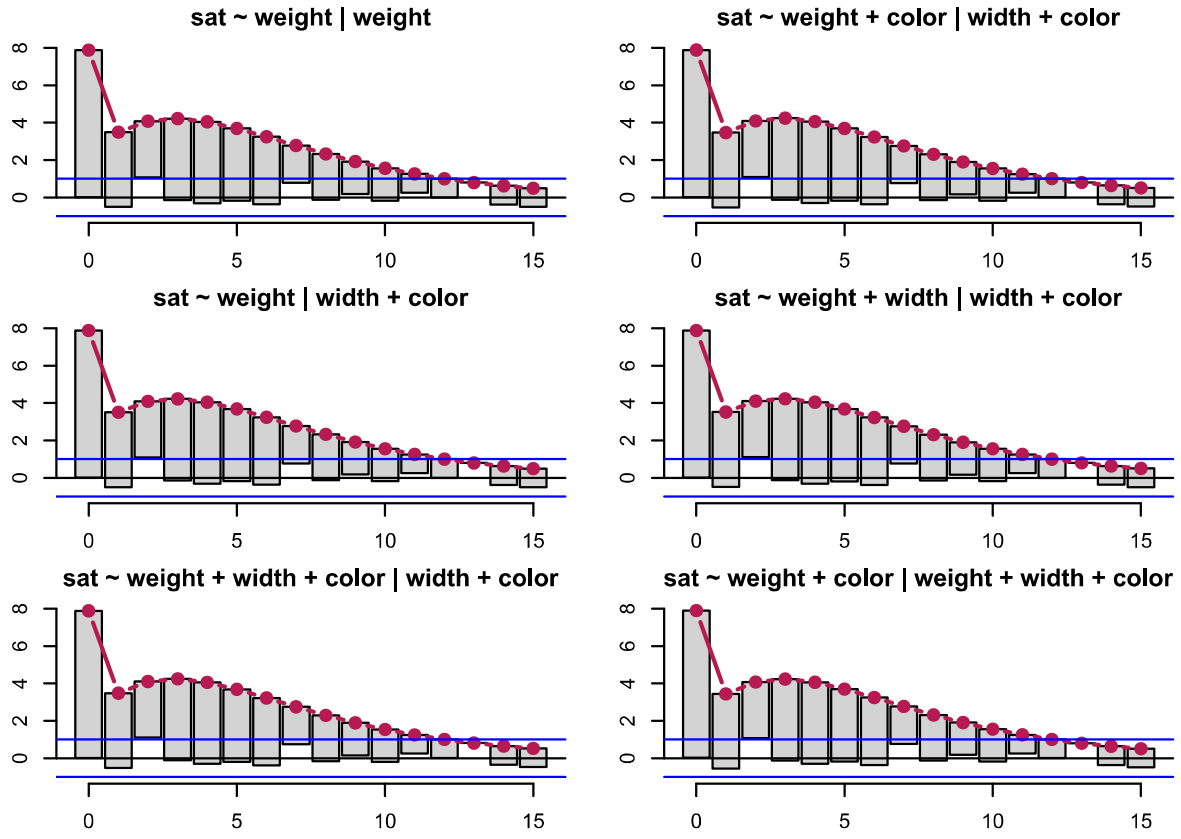


Figure 6 – Comparison of rootogram plots for best ZINB models with respect to AIC.

x-axis : Number of satellites

y-axis : $\sqrt{\text{Expected Frequencies}}$

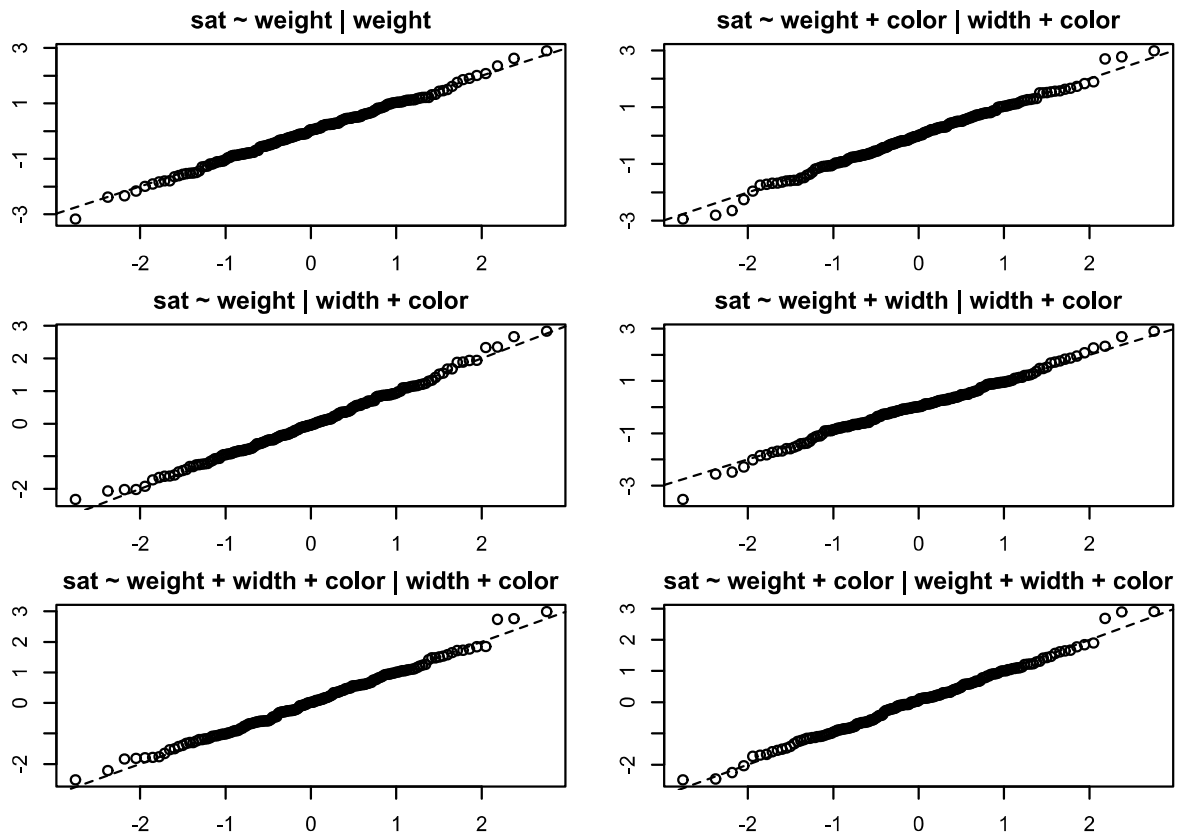


Figure 7 – Normal QQ-plot of randomized quantile residuals for best ZINB models with respect to AIC.

x-axis : Theoretical quantiles

y-axis : Empirical quantiles

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